Aseptic Technique Procedure

1. Applicability

Site	Operational Area	Applicable to
Armadale Kalamunda Group	All Clinical Areas	Medical, Nursing, Midwifery. Patient Support Services

2. Purpose

It is the intent of this procedure to ensure that all clinical staff understand the principle and process of aseptic technique (AT) and practice it effectively. This procedure will provide a standardised approach to AT by which clinical staff can be educated, assessed, and monitored to ensure compliance.

Staff are deemed competent when they have successfully completed the AKG PIVC Assessment (Appendix 2) or RPBG PIVC Assessment (Appendix 4).

3. Aseptic Technique Principles

- Asepsis is the aim for all invasive clinical procedures, including the maintenance and use of invasive clinical devices.
- Asepsis is achieved by “key-part and “key-site” protection from micro-organisms transferred from the healthcare worker or the immediate environment.
- Aseptic technique needs to be efficient as well as safe; therefore, Surgical Aseptic Technique is used for complicated procedures and Standard Aseptic Technique for uncomplicated procedures.
- A risk assessment is used to determine the need for either a surgical or standard aseptic technique based on the technical difficulty of achieving asepsis.

4. Types of Aseptic Technique

	Standard AT	Surgical AT
Core Component	**Promotes asepsis**	**Ensures asepsis**
Description	▪ Technically simple ▪ Short duration less than 20 minutes ▪ Few key sites ▪ The use of critical micro field is important to protect key parts and key sites. **Examples include:** ▪ Simple wound dressing ▪ Insertion of a PIVC ▪ Insertion of urinary catheter ▪ Change of suprapubic catheter	▪ Technically complex ▪ Takes greater than 20 minutes ▪ Large open key sites or large/numerous key parts **Examples:** ▪ Surgical procedures ▪ Insertion of a CVC ▪ Large complex wound dressing

	Standard AT	Surgical AT
Aseptic field	**Use a general aseptic field and/or critical micro aseptic field when:** ▪ Key parts can easily be protected by critical micro aseptic fields and a non-touch technique. ▪ The aseptic field does not have to be managed as a key part. ▪ **For PIVC insertion** – aseptic field is always managed as key part. *Medications trays and kidney dishes regardless of whether they are reusable or single use items, must be cleaned using a detergent impregnated wipe prior to being used as a general aseptic field.*  Yellow rectangle – general aseptic field Red circle – critical micro aseptic field	**Use a critical aseptic field and critical micro aseptic field when:** ▪ Key parts/sites are large or numerous and cannot be easily protected or handled using non-touch technique. ▪ Invasive procedures require a large aseptic working area *The critical aseptic field is always managed as a key part.*  Courtesy: RPBG Medical Multimedia Design Red rectangle – critical aseptic field
Personal protective equipment (PPE)	▪ Nonsterile gloves to remove dressing. ▪ Sterile gloves if a key part must be touched or forceps not used. ▪ Other PPE as per Standard Precautions.	▪ Sterile gloves. ▪ Often full barrier precautions – sterile gown, mask, hair covering, sterile drapes
Environment	▪ Work surface must be cleaned with pre-diluted detergent solution or hospital approved disinfectant wipes and allowed to dry before and after the procedure. ▪ Bed making or cleaning activities in proximity should be avoided.	▪ Work surface must be cleaned with pre-diluted detergent solution or hospital approved disinfectant wipes and allowed to dry before and after the procedure. ▪ Staff activity strictly controlled. ▪ Environmental risks removed or avoided.

	Standard AT	Surgical AT
	▪ Aseptic field to be over one metre from hand wash basins to avoid splashing.	

5. Aseptic Technique Application

5.1 Risk Assessment

Although the principles of aseptic technique are applied to all invasive procedures, the level of practice changes depending on a risk assessment.

The risk assessment will:

- Determine type and complexity of the procedure
- Determine what are the key-parts and key-sites
- Determine whether the key-parts or key-sites need to be touched
- Determine the appropriate infection prevention and control measures to protect key-parts and key-sites

5.2 Environmental Control Measures

Prior to conducting an aseptic procedure, ensure that there are no avoidable environmental risks nearby to reduce the risk of contamination by airborne micro-organisms. Examples of environmental risks include aseptic field less than one metre away from hand wash basin, bed making, cleaning of the environment in proximity, use of commodes by other patients in a shared room.

5.3 Infection Control Measures

Hand Hygiene	• Utilise "5 Moments for hand hygiene". Additional critical moments when hand hygiene should be performed include: ▪ Before collecting equipment ▪ Before opening the packaging of sterile items ▪ Before setting up an aseptic field ▪ Immediately before donning gloves ▪ Immediately after completing the procedure and removing gloves ▪ Immediately after cleaning equipment and/or disposing of waste • Routine hand hygiene should be performed using non medicated neutral pH soap/foam hand wash and running water or alcohol-based hand rub. • When full barrier precautions are necessary, a surgical hand scrub • using an approved antimicrobial hand wash (e.g., Chlorhexidine gluconate 4% or 2%, alcohol-based hand rub solution (Skinmann) is required.
Personal Protective Equipment (PPE)	▪ **Glove use** ▪ Non-sterile gloves may be necessary to protect the clinician from blood and body fluids or exposure to toxic drugs during administration. ▪ Sterile gloves are required in all surgical aseptic procedures and any procedures where key parts and or key sites are touched directly (i.e., when a non-touch technique cannot be achieved), to minimise the risk of contamination. ▪ Selection of sterile or non-sterile gloves is also dependent upon the clinical staff competency. ▪ When preparing for the procedure clinical staff should assess their own competence and experience in performing and determine whether touching of key parts or sites is required. If touching may take place sterile gloves are required. • **Use of other PPE equipment** ▪ Other PPE (i.e., aprons or eye/facial protection) should be worn according to standard precautions to reduce the risk of blood and body fluid exposure to the clinical staff. ▪ Maximum barrier precautions may be required during invasive procedures to reduce the risk to the patient of acquiring healthcare associated infection during procedures.
• **Aseptic Field Selection and Management**	• Prior to commencing a procedure requiring aseptic technique, the clinician needs to determine the type of aseptic field required to protect key-parts and key-sites. This includes General, Critical and Micro Critical Aseptic Fields. This ensures procedure equipment is protected from micro-organisms by using a dressing pack and individual equipment covers or, for more complex procedures, the use of sterile drapes.

Non-Touch Technique	Non-touch technique is always required to maintain asepsis. This is a technique whereby the healthcare workers hands do not touch, and thereby contaminate key-parts and key-sites. Asepsis can be achieved by either: • Using a non-touch technique e.g., sterile forceps or sterile gauze • Using sterile gloves when touching key parts or key sites directly.
Waste Management	As per standard waste management
Cleaning of equipment	All reusable equipment should be thoroughly cleaned using premixed detergent solution or hospital approved disinfectant wipes and allowed to thoroughly dry.

6. Compliance monitoring

Compliance against this procedure will be monitored by the Preventing and Controlling Infections Committee.

Compliance monitoring activities relevant to this procedure include:

- Reporting of mandatory training compliance rates including:
    - Aseptic Technique eLearning
    - Aseptic Technique Demonstrated

7. Policy context

The following documents are required to give effect to this procedure:

- East Metropolitan Health Service (EMHS) Policy: Infection Prevention and Management EMHS: 34

8. Supporting information

The following documents support the implementation of this procedure:

- Armadale Kalamunda Group (AKG) Policy: Hand Hygiene Policy (AKG-POL-0187-17)
- AKG: Infection Surveillance Procedure (AKG-PRO-0206-11)
- AKG: Cleaning of Reusable Clinical Non-Invasive Equipment Procedure
- AKG: Peripheral Intravenous Cannula (PIVC) Procedure (AK0609/23)
- Management of Peripheral Intravenous Catheters Clinical Care Standard 2021

9. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.11
- Standard 3 – Action 3.12

10. References

1. Australasian College for Infection Prevention and Control (ACIPC) Aseptic Technique Policy Practice Guidelines March 2015, Full Aseptic Technique Implementation Toolkit. 2017. Available from: Aseptic Technique Resources - ACIPC - Australasian College for Infection Prevention and Control
2. Australian College of Perioperative Nurses 2023 Standards for Safe and Quality Care in the Perioperative Environment.
3. Australian Commission on Safety and Quality in Healthcare (2019, October). National Hand Hygiene Initiative Manual.
4. Aziz, A.M. (2009). Variations in aseptic technique and implications for infection control. British Journal of Nursing, 18 (1), 26-31
5. Department of Health and Human Services, Tasmanian Infection Prevention and Control Unit. Aseptic non touch technique – A guide for healthcare workers. Version 2 2015.
6. NHMRC Australian Guidelines for the Prevention and Control of Infection in Healthcare (2010)
7. Rowley, S and Clare, S (2009) Improving standards of aseptic practice through an ANTT trust-wide implementation process: a matter of prioritisation and care. Journal of Infection prevention, 10 (1) S 18-23.

11. Glossary

The following definitions are relevant to this procedure:

Term	Definition
Asepsis	The absence of microorganism that can cause infection.
Aseptic field	Controlled workspace that contains and protects procedure equipment from contamination.
Critical aseptic field	Controlled workspace that ensures asepsis during procedures e.g., central venous catheter insertion
General aseptic field	Controlled workspace that promotes asepsis e.g., simple wound dressing
Micro critical aseptic field	A small critical aseptic field used to protect key parts or sites within critical or general aseptic field, e.g., syringe caps, covers or packaging.
	Sterile components of procedure equipment that must be protected from contamination during an aseptic procedure e.g., needles, surgical instruments, rubber tops on medication and vials, bungs
	Includes insertion site of medical devices connected to the patient. It also includes areas of the patient's skin that have been prepared using an antiseptic solution prior to a procedure or intervention.
	Measures taken to ensure the healthcare worker's hands do not touch the key parts and key sites e.g., use of sterile forceps or sterile gloves.

12. Document contact

Enquiries relating to this procedure may be directed to:

Department: Infection Prevention and Management

Email: ahsinfection.controlnurses@health.wa.gov.au

13. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4.1	29/04/2024	04/12/2026	Appendix 4 added – RPBG PIVC Insertion Assessment (Adult)

14. Authorisation

This procedure has been authorised and issued by:

Authorised by	Helen Fullarton – Director Nursing and Midwifery
Authorisation date	29/04/2024
Published date	02/05/2024

Appendix 1: Procedure Sequence

a. Sequence for procedure requiring standard aseptic technique

(Adopted from ACIPC AT Implementation Toolkit)

1. Risk Assess - Standard AT is sufficient if:				
The procedure is technically simple	Short in duration (less than 20 minutes)	Clinician is experienced/competent at performing procedure	Involves few and small key sites and key parts	Key parts or sites will not be touched

2. Apply environmental controls
Ensure that there are no avoidable nearby risk factors. This might include (but is not limited to): ▪ Waste management ▪ Cleaning of the nearby environment ▪ Bed making ▪ Patient using commode ▪ Patient bed curtains across work area

3. Consider infection control components			
Hand hygiene (Clinical wash or ABHR)	Gloves (Non-sterile or sterile)	Personal Protective Equipment	General Aseptic Field

4. Prepare for Procedure
1. Perform hand hygiene. 2. Clean the tray/trolley/work surface with the hospital approved disinfectant wipe. 3. Allow tray/trolley/work surface to thoroughly dry. 4. Identify and gather equipment for procedure. Inspect packaging for damage; check sterility indicators & expiry dates and ensure any additional equipment (e.g., tourniquet) is clean. 5. If necessary, move to where the procedure will take place. 6. Perform hand hygiene. 7. Prepare general aseptic field. Open equipment using non touch technique. 8. Position and prepare patient, using gloves where appropriate to protect from potential body fluid exposure or harmful substances.

5. Perform Procedure
9. Once ready to commence procedure and required equipment prepared, remove gloves (if used in preparation for procedure) and perform hand hygiene. 10. Wear gloves if required. If its is likely key parts or key sites will be needed to be touched directly, sterile gloves MUST be used to minimise the risk of contamination.

11. Perform the procedure using non touch technique, ensuring all key parts/components are always protected. Sterile items must only be used once and disposed into waste bag. Only sterile items may come in contact key sites and sterile items must not come into contact with non-sterile items.

6. Waste Management and Cleaning of Equipment

12. On completion of the procedure, the clinician should remove their gloves (if used) and perform hand hygiene.
13. Dispose of all waste as per standard.
14. Clean and disinfect equipment with hospital approved wipe.
15. Perform hand hygiene.

b. Sequence for procedure requiring surgical aseptic technique

(Adopted from ACIPC AT Implementation Toolkit)

1. Risk Assess – Surgical AT is sufficient if:				
The procedure is technically complex	Clinician experience/competency requires touching of key sites/parts	Long in duration (longer than 20 minutes)	Involves large open key sites or large or numerous key parts	Key parts or key sites need to be touched as part of procedure

2. Apply environmental controls

Ensure no avoidable environmental risk factors nearby. This might include (but is not limited to):

- Waste management
- Cleaning of the nearby environment
- Bed making
- Patient using commode
- Patient bed curtains across work area

3. Consider infection control components				
Hand hygiene (Surgical hand scrub/rub)	Sterile Gloves	Personal Protective Equipment	Critical Aseptic Field	Maximum barrier precautions

4. Prepare for Procedure

1. Apply PPE as required (e.g., hat, eyewear & surgical mask to protect aseptic field).
2. Perform hand hygiene.
3. Clean the tray/trolley/work surface with the hospital approved disinfectant wipe.
4. Allow tray/trolley/work surface to thoroughly dry.
5. Identify and gather equipment for procedure. Inspect packaging for damage; check sterility indicators & expiry dates and ensure any additional equipment (e.g., tourniquet) is clean.
6. If necessary, move to where the procedure will take place.
7. Perform hand hygiene.
8. Open equipment using non-touch technique.
9. Position and prepare patient, using gloves where appropriate to protect from potential body fluid exposure or harmful substances.

5. Waste Management and Cleaning of Equipment

10. On completion of the procedure, the clinician should remove their gloves and perform hand hygiene.
11. Dispose of all waste as per standard.
12. Clean and disinfect equipment with hospital approved wipe.
13. Perform hand hygiene.

Appendix 2: AKG Standard Aseptic Technique Competency Assessment

Staff Name		**HE Number**	
Department		**Designation**	
TYPE OF PROCEDURE			

This tool was adopted from ACIPC AT Implementation Toolkit and has been modified for local use.

Components		Procedure	Demonstrated	Not Demonstrated	Comments
Risk Assess	1	Undertakes risk assessment of procedure and determines appropriateness of standard AT.			
Consent & Patient Identification	2	Obtain consent from patient.			
	3	Check for allergies.			
	4	Completes patient identification using three nationally recognised identifiers.			
Manage Environmental Risks	5	Manages environmental factors prior to commencing the procedure.			
Procedure Preparation	6	Clinical Staff is bare below the elbows.			
	7	**Performs hand hygiene.**			
	8	Cleans tray/trolley/work surface as standard procedure.			
	9	Allows surface to dry before use.			
	10	Gather equipment needed and inspects packaging for damage, check sterility indicators and expiry dates.			
	11	Waste bag and sharps container within reach. ***(Waste bag can be attached to the trolley using a clip).***			
	12	**Performs hand hygiene.**			
	13	Prepares general aseptic field by opening packaging and preparing equipment using a non-touch technique. Places protective cap/s on prepared syringes ***(if applicable)*** to maintain micro-critical aseptic field where required.			
Patient preparation	14	Positions and prepares patient, using gloves where appropriate to protect from potential body fluid exposure or harmful substances.			

The yellow sections are the essential steps in Aseptic Technique

Procedure	15	**Removes gloves, if used. Performs hand hygiene.**			
	16	Applies nonsterile gloves to protect from potential body fluid exposure or applies sterile gloves if key parts or keys sites are to be touched.			
	17	Performs the procedure using non touch technique, ensuring all key parts/components are always protected.			
	18	Sterile items are only used once and disposed accordingly. Only sterile items come in contact key sites and sterile items do not encounter non-sterile items.			
	19	**Removes gloves and performs hand hygiene**.			
Patient information and documentation	20	Documents procedure details in progress notes.			
	21	Advised patient on post procedure care ***(if applicable).***			
Decontamination	22	Discard all sharps into the sharp's container.			
	23	Discards all used equipment/items into the clinical waste bin.			
	24	Clean equipment/surface/tray/trolley as per standard procedure.			
	25	**Performs hand hygiene.**			

Date of Assessment	
Assessor's Name	
Assessor's Signature	
Assessment Outcome ***Total steps in yellow criteria = 17*** ***Total steps in non-yellow criteria = 8*** ***Total steps = 25***	**Criteria** *(Please encircle)* **1. All yellow criteria demonstrated: Yes No** *(If yellow criteria not met, staff should be reassessed)* **2. Overall score ≥ 80% Yes No** Total number of steps demonstrated x 100 Total number of steps (25) ****Staff will be marked as competent if criteria 1 and 2 are met (all YES) **** ☐ **Competent** ☐ **competent**

Appendix 3: Aseptic Technique Competency Assessment for Intravenous Medication Preparation and Administration

Component		Procedure	Demonstrated	Not Demonstrated	Comments
Assessment	1	Undertakes assessment of the vascular access device for appropriateness and suitability for intravenous administration.			
Preparation (Medication Room or Prep Zone)	2	**Perform effective hand hygiene**			
	3	Disinfect medication tray/kidney dish using hospital approved wipe to prepare a general aseptic field.			
	4	Allow the tray/dish to dry.			
	5	Gather items/equipment and place around tray/dish.			
	6	**Perform effective hand hygiene** *and apply nonsterile gloves, if necessary, as per medication administration policy.*			
	7	Prepare the medication ensuring that the Key-Parts are protected with non-touch technique and Micro Critical Aseptic Fields.			
	8	Remove non-sterile gloves (if used) and **perform effective hand hygiene.**			
Patient Identification (Patient Zone)	9	Completes patient identification using three nationally recognised identifiers.			
	10	Check for allergies.			
Administration (Patient Zone)	11	**Perform effective hand hygiene** *and apply nonsterile gloves, if necessary, as per medication administration policy.*			
	12	Scrub the IV hub/administration port using 70% alcohol wipe for 15 seconds creating friction using different areas of the wipe.			
	13	Allow the hub/port to thoroughly dry.			
	14	Administer the IV medication protecting Key-Parts using non-touch technique.			
	15	Dispose of sharp and used items accordingly.			
	16	Remove non-sterile gloves (if used) and **perform effective hand hygiene.**			
Decontamination	17	Disinfect tray/dish or dispose accordingly.			
	18	**Perform effective hand hygiene.**			

Remarks: Steps 8 and 11 can be omitted if in the immediate vicinity of the patient and able to proceed to step 12 without touching anything in the environment.

The steps in this assessment tool are adopted from ANNT (Aseptic Non-Touch Technique) from www.antt.org.

The yellow sections are the essential steps in Aseptic Technique.

Date of Assessment	
Name of Assessor's	
Signature of Assessor	
Total steps in yellow criteria = 13 **Total steps in non-yellow criteria = 5** **Total steps = 18**	**Criteria** (*Please encircle)* **1. All yellow criteria demonstrated: Yes No** *(If yellow criteria not met, staff should be reassessed)* **2. Overall score > 80% Yes No** Total number of steps demonstrated x 100 Total number of steps (18) ****Staff will be marked as competent if criteria 1 and 2 are met (all YES) **** ☐ **Competent** ☐ **Not competent**

Appendix 4: RPBG PIVC Insertion Assessment (Adult)

Government of **Western Australia**
East Metropolitan Health Service

Royal Perth Bentley Group

Peripheral Intravenous Cannula (PIVC) Insertion Assessment (Adult)

Name of person being assessed (PRINT): ______________________________________

Department/ Designation: ____

Instruction for assessor: variations in technique are acceptable for the procedure; however, asepsis **must** be maintained for criteria 9 – 22 to meet this standard.

	PERFORMANCE CRITERIA	Criteria Met	
		Y/N/NA	Comments
1	Establish if health care worker has confirmed the need for peripheral intravenous cannula insertion as per PIVC CPS. Meets DRIP criteria.		
2	Perform hand hygiene		
3	Clean IV trolley with ***detergent*** and perform hand hygiene.		
4	Collect supplies including PIVC pack, sterile gloves, safety cannula, protective sheet, and place on bottom of trolley/in receptacle (e.g., kidney dish). Optional: Blood collection equipment as indicated.		
5	Identify patient using 3 core patient identifiers.		
6	Explain procedure to patient and gain appropriate consent. Clean patient's arm if visibly soiled or if local anaesthetic gel/ cream has been applied		
7	Perform hand hygiene if there was patient contact		
8	***Remove*** 'blue' insertion sticker from outer packaging of PIVC pack		
9	Open PIVC pack and contents. Prepare sterile field onto top of trolley and add sterile items: IV cannula, extension set, 10mL sodium chloride 0.9% prefilled syringe & sterile gloves (blood transfer device and syringe as appropriate)		
10	Select ***appropriate insertion site*** for required therapy. Position patient's arm on protective sheet and pillow if necessary. Apply single patient use tourniquet, above the intended insertion site and locate target vein		
11	***Liberally clean the skin*** with 2% chlorhexidine in 70% alcohol swab stick, using concentric circles ensuring site for dressing is covered. Allow to air dry for 30 seconds		
12	Perform hand hygiene and ***apply sterile gloves***		
13	Fold sterile glove packaging in half and place at the back of the sterile field as an impervious barrier that can be accessed to release tourniquet (to enable asepsis to be maintained)		
14	Separate gauze pieces. Prime extension set with sterile saline (unless blood sampling required) and return to sterile field. Remove cap from extension set. Remove backing from IV dressing for ease of application		
15	Position sterile drape over hand/ arm, only sterile gloves to touch sterile drape		
16	Apply skin traction to immobilise the vein without contaminating sterile gloves		
17	Insert cannula with bevel up, at ~25^0 angle depending on depth of vein, slowly advance cannula parallel to the vein and for a further 2-3mm after a flash of blood is observed. **N.B. if failed attempt at insertion, always use a new safety cannula**		
18	Keeping stylet still, progress ***cannula off stylet*** into the vein until the hub meets the insertion site at the skin. Activate safety device.		
19	Apply digital pressure above cannula tip to occlude vein. Disconnect safety sharp and place in sharps container **N.B. sequence of tourniquet release may vary as per inserter's preference.**		
20	Using prepared sterile glove packaging to cover non-sterile tourniquet, release tourniquet		
21	Connect extension tubing and secure with IV dressing		
22	Continuing to hold IV hub. "Scrub the hub" of extension set 70% alcohol swab if contaminated, allow to air dry. Blood sample as indicated. ***Flush IV cannula*** with 0.9% sodium chloride syringe. Check patency of IV access and observe for swelling and infiltration.		
23	Slide clamp to close prior to disconnecting flush syringe		
24	Dispose of waste. Sharps into sharps receptacle. Retain date/time insertion strip on IV dressing frame		
25	Remove gloves and perform hand hygiene		
26	Label cannula with date and time of insertion using the IV dressing. Label tourniquet with patient identification sticker (single patient item).		
27	Clean trolley with detergent		
28	Perform hand hygiene		
29	Complete blue insertion sticker and place in peripheral intravenous cannula record		

Return completed competency form to RPBG.EducationSDNs@health.wa.gov.au

Authorised by: No Patient Harm Committee October 2023

POINTS TO NOTE			
	If first cannulation attempt fails: • **If sterile field breached, use a new PIVC insertion pack**		
	Do not let the IV hub become contaminated by touching it or letting it contact the patient's skin. If contamination occurs, "scrub the hub" with 70% alcohol swab		

Comments:

____
____
____
____

Please circle: Met standard Did not meet standard

Assessor's name (PRINT): ____________________ Signature: __________________ Desig_________ Date: ________

Staff member's name (PRINT): _________________ Signature: __________________ Desig_________ Date: ________

REFERENCES

1. Royal Perth Bentley Group Cannula Insertion and Management of Peripheral Intravenous (PIVC) Clinical Guideline.
2. Royal Perth Bentley Group Aseptic Technique Policy. Infection Prevention and Management Manual.
3. Government of Western Australia, Department of Health. (2023). *Insertion and management of peripheral intravenous cannulae in healthcare facilities policy.* https://www.health.wa.gov.au/~/media/Files/Corporate/Policy-Frameworks/Public-Health/Policy/Insertion-and-Management-of-Peripheral-Intravenous-Cannulae/MP38-Insertion-and-Management-of-Peripheral-Intravenous-Cannulae.pdf
4. Australian Commission on Safety and Quality in Health Care. (2021). Management of peripheral intravenous catheters: Clinical care standard. Sydney, ACSQHC. https://www.safetyandquality.gov.au/sites/default/files/2021-05/management_of_peripheral_intravenous_catheters_clinical_care_standard_-_accessible_pdf.pdf